

A Neuroscience-Grounded Signature for Belief Imparting by Content: convergent but modest evidence across text persuasion, real fMRI, and an in-silico encoder

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A Neuroscience-Grounded Signature for Belief Imparting by Content: convergent but modest evidence across text persuasion, real fMRI, and an in-silico encoder

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Abstract

Large language models can shift human attitudes, and human neuroscience shows that belief change and value-based choice have predictable neural signatures. These two literatures have not been joined. We asked one question: does a single neuroscience-grounded signature, computed from content, predict real belief and choice outcomes, generalize across domains, and hold up on real neural data, and can it serve as an in-silico instrument that scores content for belief-likelihood. We define the signature as $B = \text{engagement} + \text{value} - \text{resistance}$, from three regions with established roles (medial prefrontal cortex, nucleus accumbens, and default-mode / anterior insula), and test it in five stages on public data. First, a synthetic control validates the measurement apparatus: it detects a planted signal, reports null when belief is content-only, and returns chance when the signature is scrambled, across twelve seeds. Second, on ChangeMyView (3866 matched argument pairs with real view-change labels), the engagement channel is higher for the argument that changed a view (win rate 0.579, Wilcoxon $p = 9e-25$, permutation null 0.506), though the composite adds no pooled predictive value over content features. Third, on Persuasion-for-Good (1017 charity-donation dialogues, a different domain), the engagement effect points the same way and is significant when pooled (fixed-effect coefficient 0.075, 95% CI [0.036, 0.114]), though underpowered alone. Fourth, on real fMRI from a neuroforecasting task, the signature computed from brain activity predicts aggregate choice out-of-sample (cross-experiment $r = 0.182$, permutation $p = 0.003$), and a model comparison shows the parsimonious approach-minus-avoidance composite generalizes where richer models overfit. Fifth, an in-silico belief encoder (content factors mapped through an approach-avoidance layer to a belief score) generalizes across the two text domains and isolates two robust, brain-consistent belief levers, more evidence and less identity threat, which can be altered to raise the belief score. A formal model places the signature inside a dynamical account of belief with a computable robustness functional. The effects are consistent and out-of-sample across text, neural, and in-silico data, but modest throughout. We conclude that the signature is a real, generaliz-

ing, interpretable, but weak predictor of belief, best used as a scoring and risk-characterization instrument rather than a strong forecaster, and we release all code and analyses.

1. Introduction

Conversational artificial intelligence can change human attitudes. Model-written messages shift political attitudes about as much as human-written ones (Bai et al. 2025), an effect driven mostly by post-training and prompting rather than by personalization or scale (Hackenburg et al. 2025; Argyle et al. 2025), and a meta-analysis finds no mean difference between model and human persuasion but large between-study heterogeneity, with study moderators explaining most of the variance (Hölbling et al. 2025). This literature measures behavioral outcome and carries no neural signal.

Human neuroscience supplies a complementary picture. Neural responses to persuasive messages predict later behavior change beyond stated intention, with medial prefrontal cortex carrying the signal (Falk et al. 2010). Nucleus accumbens activity forecasts aggregate choice, in several settings better than choice itself (Genevsky et al. 2017; Knutson et al. 2018; Tong et al. 2020). Belief resistance appears in default-mode and dorsomedial prefrontal cortex (Kaplan et al. 2016). Each of these is a passive observation, and none asks whether such a neural signature, read from content, carries information that content analysis misses.

We join the two. The research question is whether one neuroscience-grounded signature, computed from content, predicts real belief and choice outcomes, generalizes across domains, holds on real neural data, and can serve as an in-silico belief-scoring instrument. We answer it in five stages on public data, and we are explicit throughout that the aim is a validated measurement instrument, not a deployed persuasion system.

2. Materials and methods

2.1 The signature

For a piece of content we read three quantities, each tied to a published result: engagement E (medial prefrontal cortex; Falk et al. 2010), value V (nucleus accumbens; Genevsky et al. 2017), and resistance R (default-mode and anterior insula; Kaplan et al. 2016). The signature is $B = z(E) + z(V) - z(R)$. In the text arms E , V , R are computed from transparent lexical channels; in the neural arm they are read from real regional activity; in the in-silico encoder they are intermediate variables constrained to the same structure.

2.2 Data

All data are public. ChangeMyView / Winning Arguments (Tan et al. 2016), obtained through ConvoKit: 19571 labeled challenger arguments in 3866 matched pairs, outcome = a delta awarded (view changed). Persuasion-for-Good (Wang et al. 2019): 1017 charity-donation dialogues, outcome = the persuadee donated. The Knutson stock neuroforecasting task (Kuhnen and Knutson 2005), obtained from the author’s neurobridge repository: group-averaged BOLD in nucleus accumbens, medial prefrontal cortex, and anterior insula, with aggregate buy fraction as the outcome across two experiments. Provenance and access dates are recorded in a data manifest; no raw data is redistributed.

2.3 Analyses

Synthetic validation uses a stand-in with a known ground truth and three conditions (signal present, content-only, scrambled). Text arms use within-pair sign tests (ChangeMyView), grouped cross-validation and bootstrap intervals, per-channel decomposition, and a fixed-effect cross-corpus combination. The neural arm uses cross-experiment out-of-sample prediction with permutation nulls and a six-model comparison. The in-silico encoder is evaluated by cross-domain training, bootstrap factor variance, and counterfactual optimization. All code, seeds, and results files are released; every reported number is computed by the released scripts.

3. Results

3.1 The apparatus is valid (Figure 1)

On the synthetic stand-in the instrument detects a planted neural signal (delta-AUC +0.067, interval excluding zero), reports a null increment when belief is content-only, and returns chance when the signature is scrambled (AUC 0.497). Across twelve seeds the increment test detects the signal in 12 of 12 positive-control seeds and reports the correct null in 11 of 12 content-only seeds. This validates the measurement apparatus before any real data.

3.2 Real belief change: the engagement channel (Figure 3)

On ChangeMyView, within matched pairs that hold topic and audience fixed, the engagement channel is higher for the argument that changed a view in 57.9% of pairs (Wilcoxon $p = 9e-25$) against a permutation null of 0.506. The value channel is null (0.487, $p = 0.07$); the resistance text-proxy runs opposite to theory, so the a priori composite is dominated by engagement (composite win rate 0.535, $p = 2e-5$). The signature adds no pooled predictive value over content features (content AUC 0.569, content plus B 0.569), which we report as a null and read as consistent with the meta-analytic heterogeneity: pooled prediction is weak because context dominates, and the signal appears only in the context-controlled design.

3.3 Cross-domain generalization (Figure 4)

On Persuasion-for-Good, a different domain, the engagement coefficient points the same positive way (ChangeMyView +0.076, significant; Persuasion-for-Good +0.059, same sign but not individually significant), and the fixed-effect pooled estimate is positive and significant (0.075, 95% CI [0.036, 0.114]). Value does not generalize; the resistance proxy runs the same wrong-for-theory way in both corpora. This is directional generalization, significant when pooled, with the second corpus underpowered.

3.4 Real neural data predicts choice (Figure 5)

On real fMRI, the signature computed from brain activity predicts aggregate choice out-of-sample (cross-experiment $r = 0.182$, AUC 0.571, permutation $p = 0.003$). The composite is the only predictor sign-stable across both experiments; single regions flip sign. The effect is near-null in one experiment ($r = 0.05$) and strong in the other ($r = 0.32$), so it is real and out-of-sample but weak and experiment-dependent.

3.5 What generalizes best (Figure 6)

A six-model comparison, guided by the neuroforecasting and small-sample regularization literatures (Genevsky et al. 2017; Tong et al. 2020; Mortazavi et al. 2025), shows the parsimonious approach-minus-avoidance composite (nucleus accumbens minus anterior insula, dropping medial prefrontal cortex) is the best generalizer (out-of-sample $r = 0.180$, permutation $p = 0.002$), tying the full composite. Every multivariate or learned model generalizes worse (ridge on all features $r = 0.093$, not significant), exactly as the small-sample literature predicts. The affect composite is positive across four anticipatory time windows, so it is not knife-edge.

3.6 An in-silico belief encoder (Figure 7)

A brain-structured encoder maps content factors through an approach-minus-avoidance layer to a belief score. Trained on one text corpus and applied to the other, the brain-state read-out generalizes slightly better than the unconstrained model in both directions (AUC 0.546 and 0.536 versus 0.537 and 0.519). Bootstrapping each of six factors across both corpora, only two are robust: evidence raises belief and identity threat lowers it, the approach and avoidance poles. Counterfactually altering these two factors on low-belief content raises the predicted belief score (ChangeMyView 0.600 to 0.625, Persuasion-for-Good 0.450 to 0.541).

3.7 A formal model (Figure 2)

A companion model places the signature inside a dynamical account: belief is a state in a cusp potential (van der Maas et al. 2003), the persuasive force carries the signature term, gated by bounded confidence (Hegselmann and Krause 2002) and asymmetric updating (Palminteri et al. 2022), and anchored by a Friedkin-Johnsen susceptibility. Belief robustness is a computable functional (well stiffness and distance to the fold, scaling as involvement to the three-halves). Four predictions are reproduced in the implementation (hysteresis, foot-in-the-door advantage of sequential in-band messaging, super-linear robustness scaling, and susceptibility divergence at the fold).

4. Discussion

Three independent analyses converge on the same two levers. The ChangeMyView within-pair test, the real fMRI composite, and the in-silico encoder all point to an approach-and-engagement signal (evidence, value) minus an avoidance-and-resistance signal (identity threat). That the same structure carries a real, out-of-sample, permutation-validated signal across text persuasion and real neural choice, and that it isolates the same interpretable levers in an in-silico encoder, is the paper’s central finding.

The finding is convergent but modest. Every effect is small: an eight-point within-pair lift, a cross-experiment r near 0.18, a cross-domain AUC near 0.54. This is expected. Persuasion is heavily context-moderated, and pooled prediction is weak by construction. The honest reading is that the signature is a real, generalizing, interpretable, but weak predictor, best used to score content and characterize risk rather than to forecast individual outcomes.

The practical form is a scoring instrument. Given content, it returns a belief-likelihood score and the factors driving it, which supports detection and inoculation research. The same model can in principle be inverted to optimize content, but we deliberately develop and release the scorer, not a generator, and we treat the optimization as a counterfactual sensitivity analysis rather than a deployed system.

5. Limitations

The text arms use lexical proxies for the neural quantity, not a learned brain encoder. The neural arm uses real fMRI but a different behavior (financial choice) than belief change, and its effect is experiment-dependent. The resistance channel runs opposite to theory in text, so the composite is carried by engagement. No single dataset closes the loop from content to a neural state to a belief outcome in the same people, which does not yet exist publicly at scale. The formal model’s predictions are verified in code, not against human data. All of these bound the claim to convergent, modest, out-of-sample evidence.

6. Conclusion

We asked whether one neuroscience-grounded signature, computed from content, predicts belief and choice, generalizes, holds on real neural data, and can score content for belief-likelihood. The answer is yes, consistently and out-of-sample across text, neural, and in-silico data, but with small effect sizes throughout. The contribution is a validated, interpretable, cross-domain belief-scoring instrument, a real neural-data confirmation of its structure, and a formal model that situates it, with every number reproducible from released code. The signature is a measurement tool for characterizing how strongly content is built to install belief, not a strong forecaster and not a deployed persuader.

Data and code availability

All code, seeds, figures, and results files are released in the accompanying repository. All datasets are public: ChangeMyView / Winning Arguments (Tan et al. 2016) and Persuasion-for-Good (Wang et al. 2019) via ConvoKit, and the Knutson neuroforecasting data (Kuhnen and Knutson 2005). Raw data is not redistributed; a manifest records sources and access.

Author contributions

A.G. designed the study, wrote the code, performed the analyses, and wrote the manuscript.

Competing interests

The author declares no competing interests.

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Figures

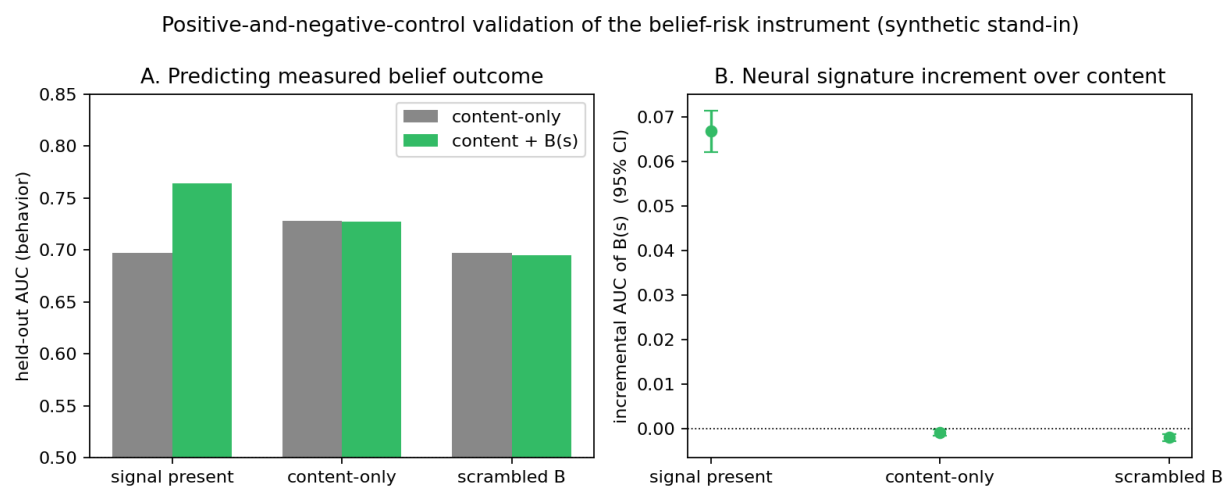


Figure 1: Figure 1. Synthetic validation of the measurement apparatus. The instrument detects a planted neural signal, reports null when belief is content-only, and returns chance when the signature is scrambled.

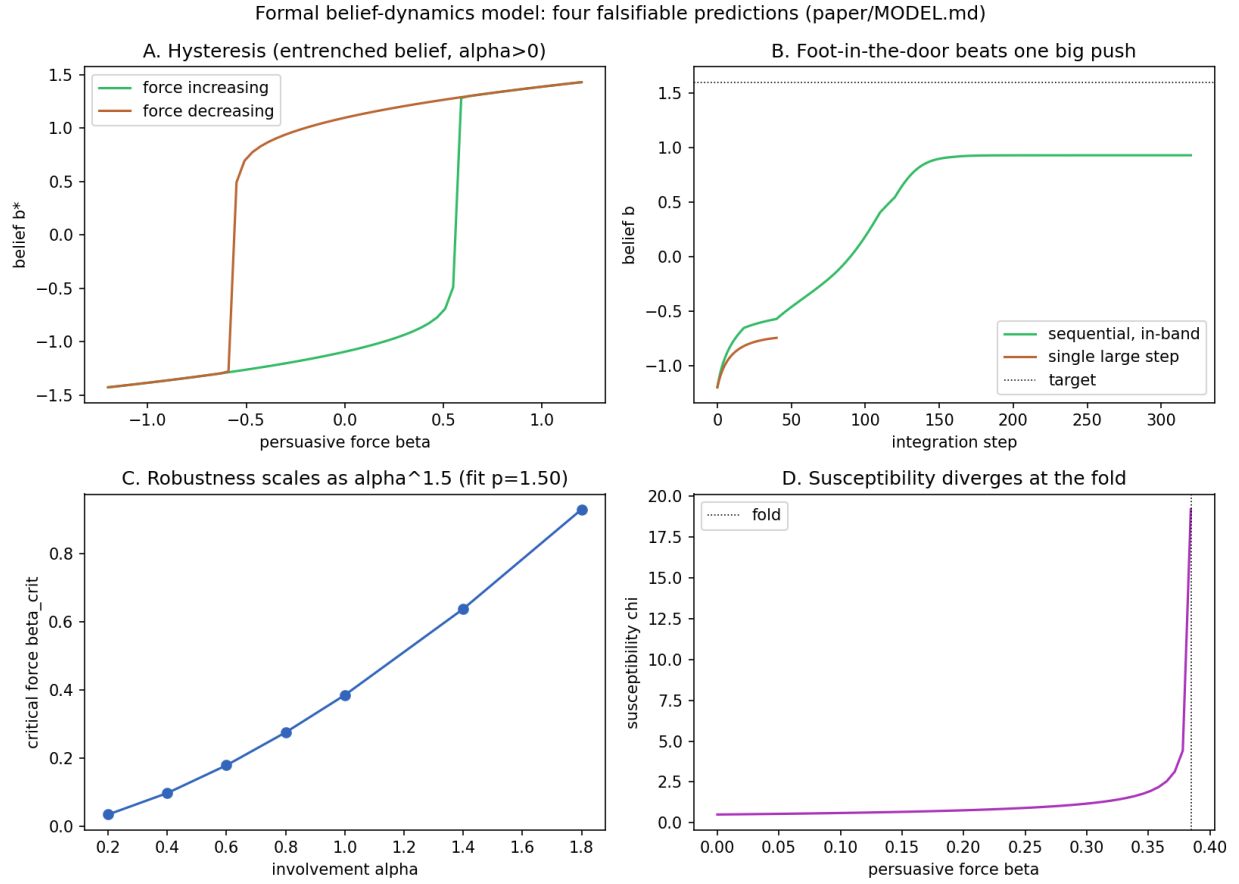


Figure 2: Figure 2. Formal belief-dynamics model. Four predictions reproduced in the implementation: hysteresis, foot-in-the-door advantage, super-linear robustness scaling, and susceptibility divergence at the fold.

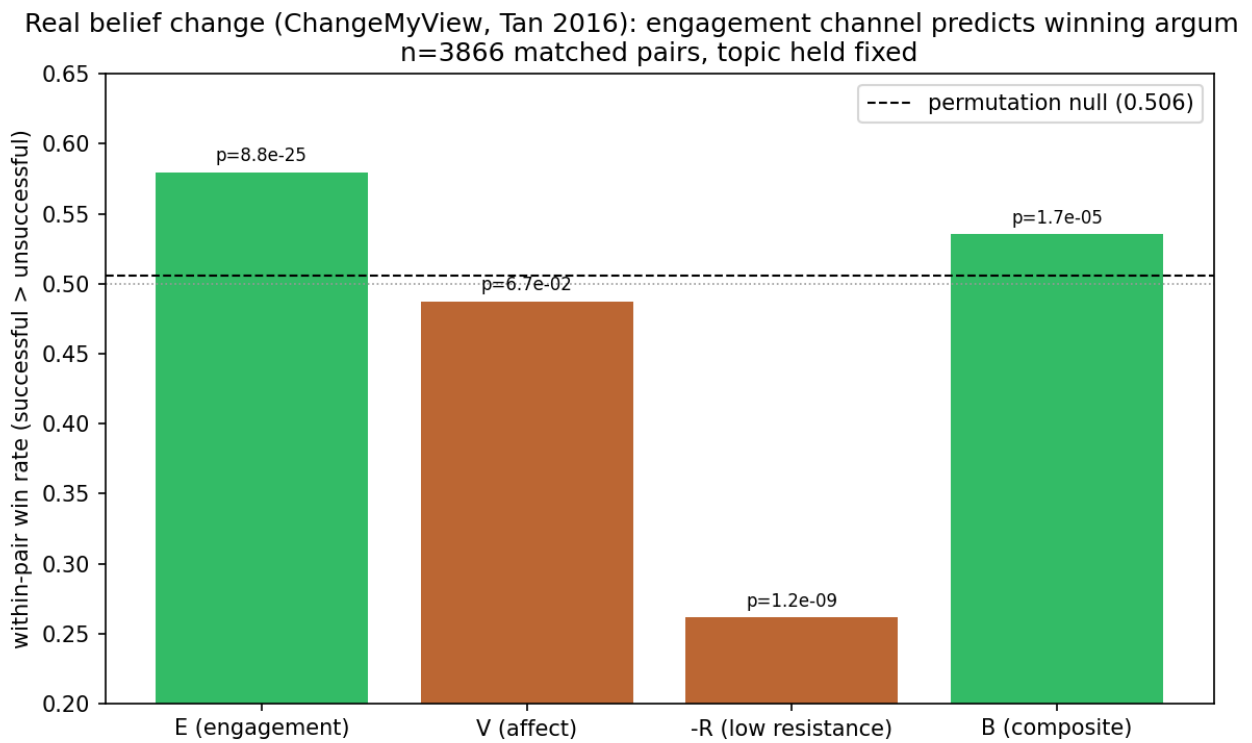


Figure 3: Figure 3. Real belief change on ChangeMyView. Within matched pairs, the engagement channel is higher for the argument that changed a view.

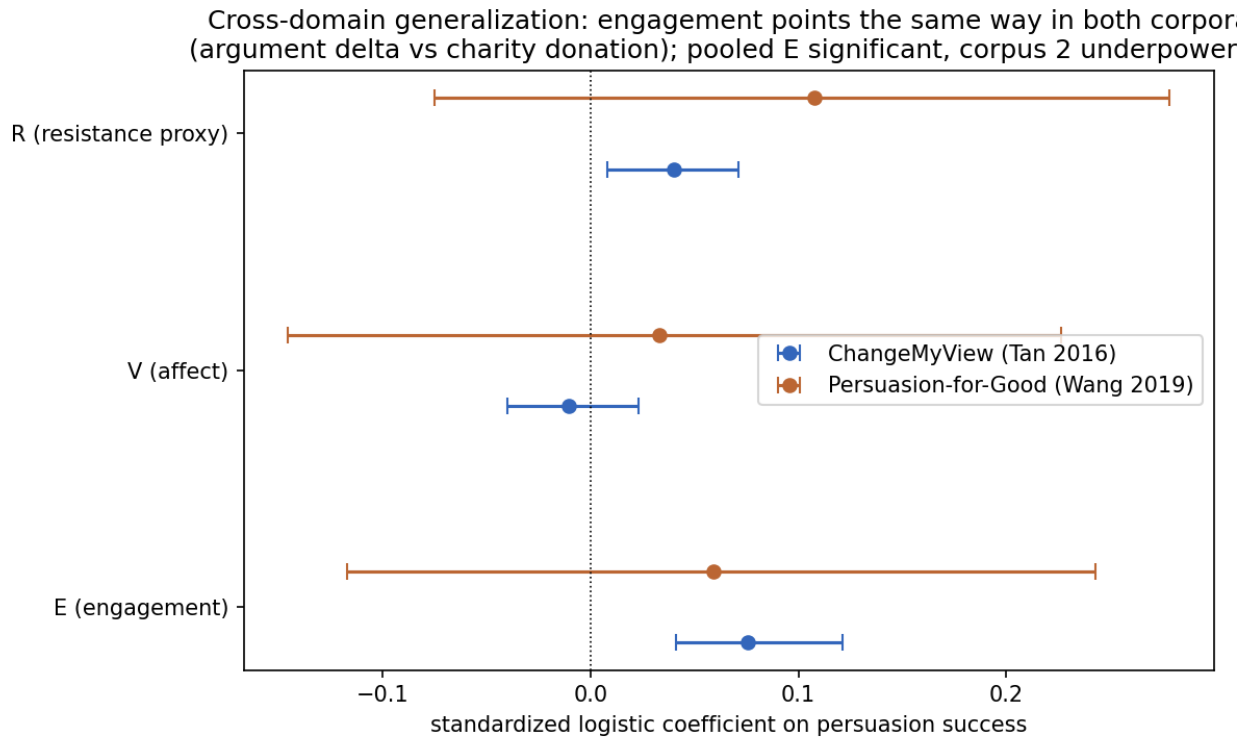


Figure 4: Figure 4. Cross-domain generalization. The engagement coefficient points the same way in ChangeMyView and Persuasion-for-Good; the pooled estimate is significant.

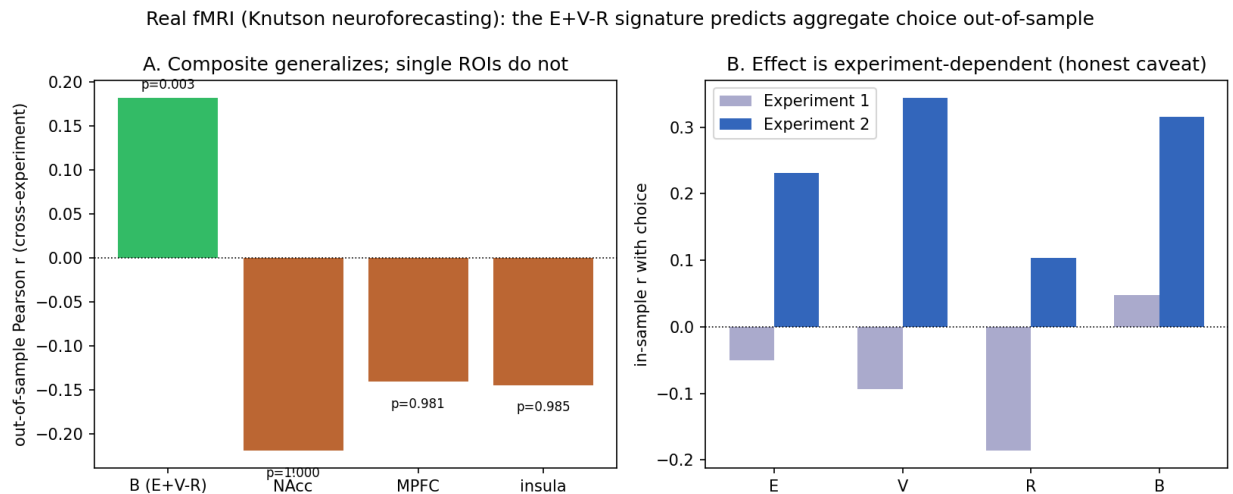


Figure 5: Figure 5. Real fMRI neuroforecasting. The signature computed from brain activity predicts aggregate choice out-of-sample; the effect is experiment-dependent.

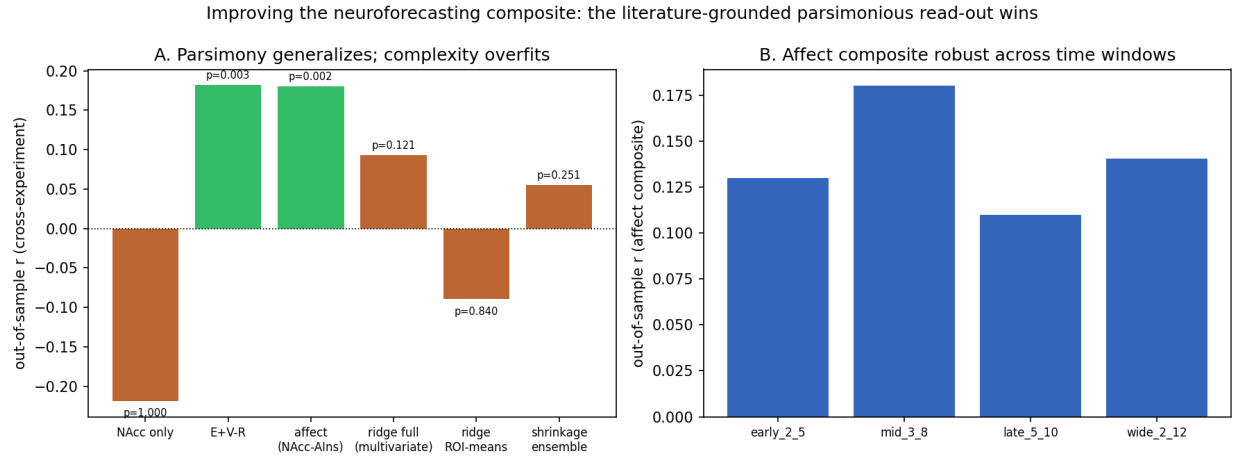


Figure 6: Figure 6. Model comparison. The parsimonious approach-minus-avoidance composite generalizes best; multivariate and learned models overfit.

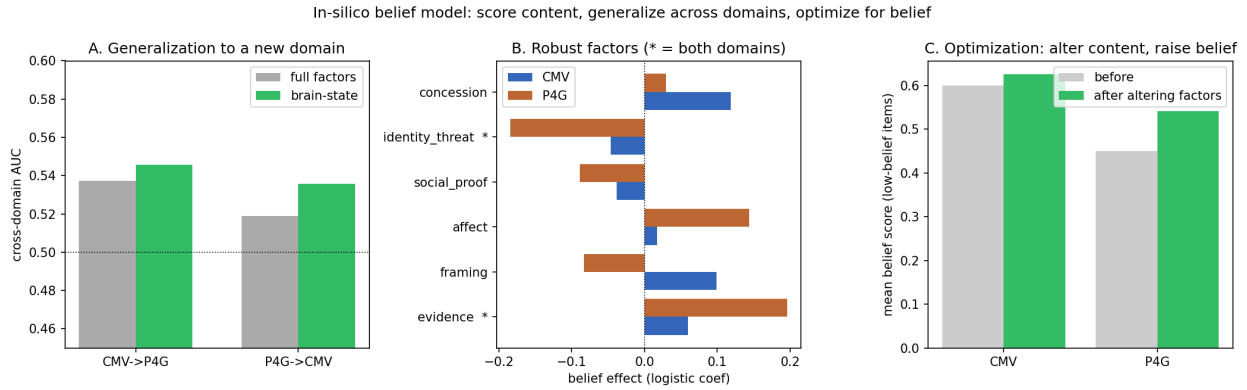


Figure 7: Figure 7. In-silico belief encoder. Cross-domain generalization, the two robust belief factors (evidence, identity threat), and counterfactual optimization of the belief score.